MTH 451/551
Lecture 7

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1 Building Polynomials

[Textbook reference: §3.1]

Many of the methods we have seen so far have been based on the idea of approximating a function by a polynomial. For example, the midpoint rule, and forward, backward, and centered difference can all be seen as approximating f by its secant. Similarly, regula falsi, Newton's method, and secant method all use linear (tangent or secant) approximations to f. These are just a few of the many important examples of applications of **polynomial approximation**. Given these, it is important to understand how we can build polynomial approximants to given functions with certain properties.

Let $n \geq 0$. Let $\mathbb{P}_n$ denote the space of polynomials of degree at most n. Note, this is the ***space*** of polynomials because this forms a vector space under the usual addition and scalar multiplication operations. The dimension of this space is $n+1$. The **monomial basis** (or *standard basis*) for $\mathbb{P}_n$ is $\{1, x, x^2, \cdots, x^n\}$. The *standard form* of a polynomial is expressed in terms of this basis,

$$p(x) = \sum_{k=0}^{n} a_k x^k = a_0 + a_1 x + \cdots + a_n x^n.$$

The largest k such that $a_k \neq 0$ is called the **leading coefficient** of p, and we write $\deg(p) = k$.

Definition 1 (Root multiplicity; Textbook Definition 3.1)**.** *If $p(x) = (x-r)^k q(x)$ for $k \in \mathbb{N}$ and some polynomial q satisfying $q(r) \neq 0$, we say that r is a **root of multiplicity** k of p. When r is a root of multiplicity 1, it is called a **simple root**.*

It can be shown that r is a root of multiplicity k of a polynomial p if and only if $p^{(j)}(r) = 0$ for $0 \leq j \leq k-1$, but $p^{(k)}(r) \neq 0$.

Theorem 1 (Fundamental theorem of algebra and consequences; Textbook Theorem 3.2)**.** *Let $p \in \mathbb{P}_n$ for some $n \geq 1$. The following hold:*

1. *If* $\deg(p) = n$ *(so* $a_n \neq 0$*), then* p *has precisely* n *roots in* $\mathbb{C}$*,* $r_1, \dots, r_n$*, counting multiplicities. Furthermore,*
$$p(x) = a_n \prod_{k=1}^{n} (x - r_k) = a_n (x - r_1) \cdots (x - r_n).$$

2. *If* p *has more than* n *roots, counting multiplicities, then* $p = 0$*.*

Although the monomial basis is the one we are most familiar with, there are other bases that may be (much) more useful in actual computations. To give one motivating example, we discuss **interpolating polynomials**. A function f **interpolates** points (x_i, y_i) if it satisfies $f(x_i) = y_i$. For a collection of distinct points x_i and values y_i, for $i = 0, 1, \dots, n$, we can construct an **interpolating polynomial**,
$$p(x) = \sum_{j=0}^{n} \left(y_j \prod_{\substack{k=0 \\ k \neq j}}^{n} \frac{x - x_k}{x_j - x_k} \right).$$
The interpolation property can be verified by plugging in x_j, and noticing that all but one term in the sum vanish. In this representation, y_j are the coefficients of the so-called **Lagrange basis functions**.

- **Shifted monomial basis.** Given a number c, the shifted monomial basis is $\{1, x - c, (x - c)^2, \dots, (x - c)^n\}$. This is the basis used for Taylor polynomials.
- **Lagrange basis.** Given distinct numbers $\{x_0, \dots, x_n\}$, the Lagrange polynomials are
$$\ell_j(x) = \prod_{\substack{k=0 \\ k \neq j}}^{n} \frac{x - x_k}{x_j - x_k}.$$
Each of these polynomials is of degree n, and they satisfy the *Kronecker delta property*,
$$\ell_j(x_i) = \delta_{ij} := \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$
Note that $\sum_{j=0}^{n} \ell_j(x) = 1$.
- **Newton basis.** Similar to the Lagrange basis, the **Newton polynomials** are given by
$$w_j(x) = \prod_{k=0}^{j-1} (x - x_k).$$
Note that $\deg(w_j) = j$ and $w_j(x_i) = 0$ when $j > i$.

1.1 Constraints and Functionals

Since $\mathbb{P}_n$ is an $(n+1)$-dimensional vector space, we expect that an element of $\mathbb{P}_{n+1}$ can be determined by $n+1$ constraints. We will soon given conditions on a set of $n+1$ constraints that will guarantee that they will determine a unique polynomial. This will rely on the concept of a **linear functional**.

Definition 2. *A **linear functional** on a vector space V is a function $\phi : V \to \mathbb{R}$ satisfying*

$$\phi(u+v) = \phi(u) + \phi(v), \qquad \text{and} \qquad \phi(cu) = c\phi(u),$$

for all $u, v \in V$ and $c \in \mathbb{R}$.

In other words, a functional is a linear operator from V to $\mathbb{R}$. As two examples of linear functionals on $\mathbb{P}_n$:

- **Evaluation at a point.** For $c \in \mathbb{R}$ and integer $j \geq 0$, $\phi(p) = p^{(j)}(c)$.
- **Integration on an interval.** For $a < b$ and an integer $j \geq 0$, $\phi(p) = \int_a^b p(x)x^j\,dx$.

We can verify that these are linear functionals by directly checking the definition and using linearity of the derivative and integral.

Lemma 1 (Unique determination of $p \in \mathbb{P}_n$; Textbook Lemma 3.4)**.** *Suppose that $\phi_j : \mathbb{P}_n \to \mathbb{R}$, $0 \leq j \leq n$, are functionals. The following are equivalent:*

1. *The only $p \in \mathbb{P}_n$ for which $\phi_0(p) = \phi_1(p) = \cdots = \phi_n(p) = 0$ is $p = 0$.*
2. *For each set of numbers $\{y_0, \ldots, y_n\}$, there is a unique polynomial $p \in \mathbb{P}_n$ satisfying $\phi_j(p) = y_j$ for each $0 \leq j \leq n$.*

Proof. Let $\Phi : \mathbb{P}_n \to \mathbb{R}^{n+1}$ be the linear transformation given by $\Phi(p) = (\phi_0(p), \ldots, \phi_n(p))$. The range and nullspace of Φ are, respectively, $\mathscr{R}(\Phi) = \{\Phi(p) : p \in \mathbb{P}_n\} \subset \mathbb{R}^{n+1}$ and $\mathscr{N}(\Phi) = \{p \in \mathbb{P}_n : \Phi(p) = 0\} \subset \mathbb{P}_n$. By the Rank-Nullity Theorem, $\dim \mathscr{R}(\Phi) + \dim \mathscr{N}(\Phi) = \dim \mathbb{P}_n = n+1$. The first condition is the same as saying that $\mathscr{N}(\Phi) = \{0\}$, from which we determine that $\dim \mathscr{R}(\Phi) = n+1$. Since $\dim(\mathbb{R}^{n+1}) = n+1$, this implies that $\mathscr{R}(\Phi) = \mathbb{R}^{n+1}$, which is equivalent to the second condition. Conversely, the second condition is the same as saying that $\text{Range}(\Phi) = \mathbb{R}^{n+1}$, from which we determine that $\dim(\text{Nullspace}(\Phi)) = 0$, i.e. $\text{Nullspace}(\Phi) = \{0\}$, which is equivalent to the first condition. □

We give a second proof, which also provides a means of constructing the desired polynomial. If we take $(b_0, b_1, \ldots, b_n)$ to be a basis for $\mathbb{P}_n$, we can recast both conditions above in terms of matrix algebra. Any $p \in \mathbb{P}_n$ is uniquely identified with a coefficient vector $c = (c_0, \ldots, c_n) \in \mathbb{R}^{n+1}$ by its unique representation with respect to the basis, $p(x) = c_0 b_0(x) + \cdots + c_n b_n(x)$. Letting $A \in \mathbb{R}^{(n+1)\times(n+1)}$ be defined by $a_{ij} = \phi_i(b_j)$ for $0 \le i, j \le n$, the two conditions can be rephrased as follows:

1. The only $c \in \mathbb{R}^{n+1}$ for which $Ac = 0$ is $c = 0$.
2. For each $y \in \mathbb{R}^{n+1}$, there is a unique $c \in \mathbb{R}^{n+1}$ such that $Ac = y$.

Either of these two statements (conditions) imply the other, and both imply that A is invertible. Using $c = A^{-1}y$, we can construct p in terms of the basis.

The Fundamental Theorem of Algebra provides the standard tool for checking Condition 1 of Lemma 1 for a set of functionals that employ evaluation at a point.

Theorem 2 (Interpolating polynomial; Textbook Theorem 3.5)**.** *Suppose that $\{x_0, \ldots, x_n\}$ are distinct. For each set of numbers $\{y_0, \ldots, y_n\}$, there is a unique polynomial $p \in \mathbb{P}_n$ satisfying $p(x_j) = y_j$ for each j.*

Proof. We define the functionals ϕ_j by $\phi_j(p) = p(x_j)$ for $p \in \mathbb{P}_n$. Suppose that $p \in \mathbb{P}_n$, and $\phi_j(p) = 0$ for each j. This means that p has at least $n + 1$ distinct roots, one at each x_j. But the degree of p is at most n, so the only way that it can have more than n distinct roots is if it is the constant polynomial $p = 0$ (by Theorem 1). Now we apply Lemma 1 to complete the proof. □

Example 1 (Vandermonde matrix; Textbook Example 3.6)**.** If we were to give an interpolating polynomial in terms of the monomial basis, $p(x) = \sum_{j=0}^{n} a_j x^j$, the interpolation conditions, $p(x_i) = y_i$ for $0 \le i \le n$, would naturally lead to an $(n+1)\times(n+1)$ linear system involving a **Vandermonde matrix**

$$\begin{pmatrix} 1 & x_0 & \cdots & x_0^n \\ 1 & x_1 & \cdots & x_1^n \\ \vdots & \vdots & & \vdots \\ 1 & x_n & \cdots & x_n^n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{pmatrix}.$$

Similarly, using the shifted monomial basis, $p(x) = \sum_{j=0}^{n} a_j (x - c)^j$, we obtain the linear system

$$\begin{pmatrix} 1 & x_0 - c & \cdots & (x_0 - c)^n \\ 1 & x_1 - c & \cdots & (x_1 - c)^n \\ \vdots & \vdots & & \vdots \\ 1 & x_n - c & \cdots & (x_n - c)^n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{pmatrix}.$$

This approach is rarely used for practical numerical computations where interpolation is the focus, but it can be helpful in deriving finite difference approximations of derivatives of a function.

Theorem 2 implies that the Vandermonde matrix is invertible. It is an exercise to show that, for any numbers $\{z_0, z_1, \dots, z_n\}$, the corresponding Vandermonde determinant is given by

$$\begin{vmatrix} 1 & z_0 & \cdots & z_0^n \\ 1 & z_1 & \cdots & z_1^n \\ \vdots & \vdots & & \vdots \\ 1 & z_n & \cdots & z_n^n \end{vmatrix} = \prod_{i<j} (z_j - z_i) = \prod_{i=0}^{n-1} \prod_{j=i+1}^{n} (z_j - z_i).$$

If the numbers are distinct, this determinant is clearly non-zero. ⧫

Example 2 (Newton basis; Textbook Example 3.6)**.** If the interpolating polynomial is given in the Newton basis, $p(x) = \sum_{j=0}^{n} c_j w_j(x)$, the associated linear system is

$$\begin{pmatrix} w_0(x_0) & & & \\ w_0(x_1) & w_1(x_1) & & \\ \vdots & \vdots & \ddots & \\ w_0(x_n) & w_1(x_n) & \cdots & w_n(x_n) \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{pmatrix}.$$

We note that the first column of this matrix consists only of ones, since $w_0(x) = 1$. The empty space above the diagonal entries indicates that these entries are zero. This is because, for $j > i$, the product

$$w_j(x_i) = \prod_{k=0}^{j-1} (x_i - x_k)$$

contains the factor $(x_i - x_i) = 0$ (since $0 \le i \le j - 1$), so $w_j(x_i) = 0$ whenever $j > i$. Since the matrix is lower-triangular, it is straightforward to solve the system by substitution, but we will later see that the notion of divided differences provides a more efficient way of computing the coefficients. ⧫

Theorem 3 (Taylor polynomial; Textbook Theorem 3.7)**.** *Let $c \in \mathbb{R}$. For each set of numbers $\{y_0, \dots, y_n\}$, there is a unique polynomial $p \in \mathbb{P}_n$ satisfying $p^{(j)}(c) = y_j$ for each j.*

Proof. We define the functionals ϕ_j by $\phi_j(p) = p^{(j)}(c)$ for $p \in \mathbb{P}_n$. Suppose that $p \in \mathbb{P}_n$, and $\phi_j(p) = 0$ for each j. This means that p has a root of multiplicity at least $n + 1$ at c. But the degree of p is at most n, so it must be the constant polynomial $p = 0$ (by Theorem 1). Now we apply Lemma 1 to complete the proof. In this case, we can verify directly that

$$p(x) = \sum_{j=0}^{n} \frac{y_j}{j!} (x - c)^j$$

is this unique polynomial. □

In particular, if $y_j = f^{(j)}(c)$ for some function f, then this unique polynomial is exactly the degree-n Taylor polynomial of f centered at c.

Example 3 (Vandermonde inverse; Textbook Example 3.8)**.** Let $\{x_0, \dots, x_n\}$ be distinct numbers, and let $\ell_j \in \mathbb{P}_n$, $0 \le j \le n$, be the associated Lagrange polynomials. Let c be another number, which may or may not be one of the x_j. According to Theorem 3, we can re-express each Lagrange polynomial in the shifted monomial basis as

$$\ell_j(x) = \sum_{k=0}^{n} \frac{\ell_j^{(k)}(c)}{k!}(x-c)^k.$$

This simple change of basis tells us something useful about the inverse of the Vandermonde matrix,

$$V = \begin{pmatrix} 1 & x_0 - c & \cdots & (x_0-c)^n \\ 1 & x_1 - c & \cdots & (x_1-c)^n \\ \vdots & \vdots & & \vdots \\ 1 & x_n - c & \cdots & (x_n-c)^n \end{pmatrix}.$$

Specifically, if we define the matrix U by $u_{ij} = \ell_j^{(i)}(c)/i!$ for $0 \le i, j \le n$, then the (i, j) entry of VU is

$$[VU]_{ij} = \sum_{k=0}^{n} (x_i - c)^k \frac{\ell_j^{(k)}(c)}{k!} = \ell_j(x_i) = \delta_{ij}.$$

This tells us that $VU = I$, so $U = V^{-1}$, i.e.

$$V^{-1} = \begin{pmatrix} \ell_0(c) & \ell_1(c) & \cdots & \ell_n(c) \\ \ell_0'(c) & \ell_1'(c) & \cdots & \ell_n'(c) \\ \ell_0''(c)/2 & \ell_1''(c)/2 & \cdots & \ell_n''(c)/2 \\ \vdots & \vdots & & \vdots \\ \ell_0^{(n)}(c)/n! & \ell_1^{(n)}(c)/n! & \cdots & \ell_n^{(n)}(c)/n! \end{pmatrix}.$$

This relationship will be of use later for deriving some schemes for approximating derivatives. ♦

Although most of this chapter concerns polynomial approximations for which values of the polynomial, and possibly some derivatives, are specified at given points, we will also consider polynomials that are defined in terms of integral moments, as in the following theorem.

Theorem 4 (Least-squares approximation; Textbook Theorem 3.9)**.** *Let $a, b \in \mathbb{R}$ with $a < b$. For each set of numbers $\{y_0, \dots, y_n\}$, there is a unique polynomial $p \in \mathbb{P}_n$ satisfying $\int_a^b p(x)(x-a)^j\,dx = y_j$ for each j.*

Proof. In this case, we take ϕ_j to be $\phi_j(p) = \int_a^b p(x)(x-a)^j\,dx$. Now suppose that $p \in \mathbb{P}_n$, and $\phi_j(p) = 0$ for $0 \le j \le n$. Well, $p(x) = \sum_{j=0}^n c_j(x-a)^j$ for some coefficients c_j, so we see that

$$0 = \sum_{j=0}^{n} c_j \phi_j(p) = \sum_{j=0}^{n} c_j \int_a^b p(x)(x-a)^j\,dx = \int_a^b p(x) \sum_{j=0}^{n} c_j (x-a)^j\,dx = \int_a^b [p(x)]^2\,dx.$$

Since p^2 is non-negative and continuous on $[a, b]$, the only way it can integrate to zero is if $p^2 = 0$ on $[a, b]$, which implies that $p = 0$ (everywhere). This completes the proof. □